EC991 Topics in Macroeconomics

Week 2 · The Consumption–Savings Problem

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Introduction

Goal.

- Present the canonical dynamic general equilibrium model of **incomplete markets** with **household heterogeneity**
- The framework speaks to questions such as:
  - How much wealth inequality can earnings variation across agents explain?
  - What are the distributional implications of fiscal policies? How do inequality and welfare respond?
  - What are the consequences of this heterogeneity for aggregate variables and prices?
- Focus: the **stationary equilibrium**, with prices constant through time

Introduction

Model ingredients.

- A consumption-savings problem in infinite horizon, with two features:
  1. **Idiosyncratic shocks**: exogenous "income shocks" (unemployment, promotions, ...)
  2. **Incomplete markets**: no way to buy insurance against those shocks in the market
- There is **NO aggregate uncertainty**

Ex-ante homogeneous, ex-post heterogeneous.

- **Exogenous** earnings distribution, but **endogenous** wealth distribution
- Lucky individuals, after a run of high-income shocks, accumulate assets to insure against future low income
- Unlucky individuals, after bad shocks, end up with no assets
- Equilibrium features a **non-degenerate stationary wealth distribution**

This lecture

Part 1. What "incomplete markets" takes away.

- A $2 \times 2 \times 2$ economy: two households, two dates, two states
  - with **Arrow securities**: full insurance, and the price of a state
  - with a **riskless bond** only: no insurance, and a lower interest rate
- Infinite horizon, one consumer, prices given
  - complete markets: constant consumption, assets tied to the state
  - a bond only: consumption follows a martingale, assets drift

Part 2. The Bewley household.

- Prudent households with a borrowing limit: constrained choices and saving buffers
- The recursive problem, and the step to general equilibrium

A $2 \times 2 \times 2$ economy

Two households, two dates, two states

- Households $i = 1, 2$; dates $t = 0, 1$; at date 1 a state $s \in \{A, B\}$, each with probability $\pi_s = \frac{1}{2}$
- Date 0: both households receive $\bar{y}$
- A and B label the **state of the whole economy**; H and L label income levels

Common state at date 1	Household 1 income	Household 2 income
A	y_H	y_L
B	y_L	y_H

- Same preferences and income prospects, but **perfectly negatively correlated** date-1 incomes
  - $y_H > \bar{y} > y_L$ with $\bar{y} = \frac{1}{2}(y_H + y_L)$
  - ex-ante identical, ex-post different
- Aggregate income is $2\bar{y}$ at both dates in both states: **idiosyncratic risk only**, as in Part 2
- Preferences, identical across households:

$$u(c_0^i) \; + \; \beta \sum_s \pi_s \, u\big(c_1^i(s)\big), \qquad u' > 0, \; u'' < 0$$

- How much of the date-1 risk can they share? It depends on **what they are allowed to trade**

Two market structures

Complete markets.

- Households can trade a security for **every** possible future state: two **Arrow securities**
  - security s costs $q(s)$ today and pays one unit at date 1 if state s occurs, nothing otherwise
- They can arrange to receive money when they need it, in exchange for paying when they are better off

Incomplete markets, in this example.

- Households can trade only a **riskless bond**: it costs $1/R$ today and pays one unit at date 1 in both states
- They can move resources from today to tomorrow, but cannot make the payment depend on tomorrow's income

Common to both.

- Numbers for later: $\beta = 0.96$, CRRA with $\gamma = 2$, $\bar{y} = 1$, $y_H = 1.2$, $y_L = 0.8$

Gross return

- Buy one A-security and one B-security today:

Tomorrow's outcome	A-security pays	B-security pays	Total payment
A occurs	1	0	1
B occurs	0	1	1

- This portfolio costs $q(A) + q(B)$ today and guarantees one unit tomorrow: it **is** the riskless bond
- Its gross return is therefore

$$R = \frac{\text{payment tomorrow}}{\text{cost today}} = \frac{1}{q(A) + q(B)}$$

Complete markets: the household problem

Household i chooses c_0^i and claims $a^i(A)$, $a^i(B)$:

$$\max\ u(c_0^i) + \beta \sum_s \pi_s\, u(c_1^i(s)) \quad \text{s.t.} \quad c_0^i + \sum_s q(s)\, a^i(s) = \bar{y}, \qquad c_1^i(s) = y^i(s) + a^i(s)$$

- $a^i(s) < 0$ means selling a claim on state s: a promise to pay if s occurs
- No constraint beyond the budget: $c_1^i(s) > 0$ in each state is all that is needed
- First-order condition for $a^i(s)$, one per state:

$$q(s) \;=\; \beta\, \pi_s\, \frac{u'(c_1^i(s))}{u'(c_0^i)}$$

- The price of a unit in state s: discounting, times the probability of s, times how hungry the household is in s

Complete markets: equilibrium

- Market clearing in each common state $s = A, B$: $a^1(s) + a^2(s) = 0$
  - so $c_1^1(s) + c_1^2(s) = 2\bar{y}$ in each state
- Both households face the same $q(s)$, so the first-order conditions equalise marginal-utility ratios:

$$\frac{u'(c_1^1(s))}{u'(c_0^1)} = \frac{u'(c_1^2(s))}{u'(c_0^2)} \qquad \text{for } s = A, B$$

- Could this common ratio exceed one? Then both households would consume less in state s than today
  - combined consumption in state s would fall below $2\bar{y}$, but aggregate income is $2\bar{y}$ at both dates
  - a ratio below one fails the same way, so the ratio is one: $c_1^i(s) = c_0^i$ for both households, in both states
- Identical preferences and date-0 income make the equilibrium symmetric: $c_0^i = \bar{y}$, so $c_1^i(s) = \bar{y}$
- **Full insurance**: individual consumption moves with aggregate income (constant) and not with the state
- Prices follow from the first-order condition with $c_1 = c_0$:

$$q(s) = \beta\,\pi_s, \qquad q(A) + q(B) = \beta, \qquad R = 1/\beta$$

- A more patient household gives up more consumption today for the same payment tomorrow: a higher price $q(s)$

Complete markets: what is traded

With $\bar{y} = 1$, $y_H = 1.2$, $y_L = 0.8$, $\beta = 0.96$: household 1 sells 0.2 units of the A claim and buys 0.2 units of the B claim.

State	π_s	$q(s)$	Income of household 1	$a^1(s)$	$c_1^1(s)$
A	0.5	0.48	1.20	-0.20	**1.00**
B	0.5	0.48	0.80	$+0.20$	**1.00**

- The portfolio costs nothing at date 0: $0.48 \times (-0.2) + 0.48 \times 0.2 = 0$, so $c_0 = \bar{y}$ too
- Household 2 holds the opposite position: each household **insures the other**
- Gross interest rate $R = 1/\beta$, so $r =$ **4.2%**

Incomplete markets: the household problem

Household i chooses c_0^i and a bond position a^i, the same in both states:

$$\max\ u(c_0^i) + \beta \sum_s \pi_s\, u\big(c_1^i(s)\big) \quad \text{s.t.} \quad c_0^i + \frac{a^i}{R} = \bar{y}, \qquad c_1^i(s) = y^i(s) + a^i$$

- The bond moves resources **across dates**, not **across states**: $c_1^i(A) - c_1^i(B) = y^i(A) - y^i(B)$ whatever a^i is
- One first-order condition instead of two, the **Euler equation**:

$$u'(c_0^i) \;=\; \beta R \sum_s \pi_s\, u'\big(c_1^i(s)\big) \;=\; \beta R\, \mathbb{E}\big[u'(c_1^i)\big]$$

- Households can no longer insure their income
- Does this make them want to **save more**, and what happens to the **equilibrium interest rate**?

Detour: risk aversion and prudence

Compare the income lottery $y^i(s)$ with certain income $\bar{y} = \mathbb{E}[y^i(s)]$.

Risk aversion: does the household dislike the lottery?

$$u'' < 0 \quad \Longrightarrow \quad \mathbb{E}\big[u(y^i(s))\big] < u(\bar{y})$$

- Concave utility: the loss in the bad outcome outweighs the gain in the good outcome
- The household values insurance that removes risk without reducing mean consumption

Prudence: does risk make extra future consumption more valuable?

$$u''' > 0 \quad \Longrightarrow \quad \mathbb{E}\big[u'(y^i(s))\big] > u'(\bar{y})$$

- Convex marginal utility: its rise in the bad outcome exceeds its fall in the good outcome
- CRRA ($\gamma > 0$, including log): risk averse and prudent
- Quadratic utility: risk averse, but $u''' = 0$ and $\mathbb{E}[u'(y^i(s))] = u'(\bar{y})$

Why the distinction matters.

- Risk aversion explains why you want **insurance**
- Prudence explains why uninsured risk can make you want a larger **saving buffer**

More details

Incomplete markets: what the bond can and cannot do

Keep the complete-markets return $R = 1/\beta$ and start from zero bond holdings.

1. No income risk.

- Consumption is 1 today and 1 tomorrow
- At $\beta R = 1$ the return exactly offsets impatience: zero saving is optimal, consumption is already smooth

2. Income risk tomorrow.

	Today	Tomorrow, low	Tomorrow, high
Consumption with zero saving	1	0.8	1.2
Effect of buying a bond	less	more	more

- The household would like to move consumption from high-income to low-income outcomes: that is **insurance**
- The bond cannot do that: it raises consumption in **both** states tomorrow
- Low income: consumption becomes smoother across dates. High income: it becomes less smooth
- Two opposing effects, and the household must choose before it knows the state

Incomplete markets: the desire to save

Which effect wins? Saving a little costs $u'(1)$ today and returns $\beta R\,\mathbb{E}[u'(y^i(s))]$ tomorrow. With $\beta R = 1$:

$$\mathbb{E}\big[u'(y^i(s))\big] - u'(1) \;=\; \tfrac{1}{2}\underbrace{\big[u'(0.8) - u'(1)\big]}_{\text{gain when income is low}} \;-\; \tfrac{1}{2}\underbrace{\big[u'(1) - u'(1.2)\big]}_{\text{loss when income is high}}$$

	$u'(0.8)$	$u'(1)$	$u'(1.2)$	$\mathbb{E}[u'] - u'(1)$
Quadratic, $u'(c) = 2 - c$	1.2	1	0.8	0
Log, $u'(c) = 1/c$	1.25	1	0.83	0.04

- Quadratic utility: extra consumption is worth 0.2 more when income is low and 0.2 less when income is high
  - Expected marginal utility tomorrow is 1, the same as today: zero saving stays optimal
  - The household still dislikes the risk and would buy insurance; the bond is just a poor way to reduce it
- Log utility: the low income raises u' by more than the high income lowers it: **prudence** makes saving attractive
  - Smoothing alone generates saving only if income is expected to fall; here the mean is unchanged
  - Both households want to **lend** at this return. But who will borrow? **This cannot be an equilibrium**

More details

Incomplete markets: equilibrium

The return falls until households are willing to hold zero bonds.

- The bond is in **zero net supply**: clearing requires $a^1 + a^2 = 0$
- Symmetric households choose the same position, so $a^1 = a^2 = 0$: consumption tracks income, $c_1^i(s) = y^i(s)$
- A lower R makes saving less attractive; the Euler equation gives the return at which zero saving is optimal:

$$R = \frac{u'(\bar{y})}{\beta \, \mathbb{E}\left[u'(y^i(s))\right]}$$

- With prudence, $\mathbb{E}[u'(y^i(s))] > u'(\bar{y})$, so $\beta R < 1$; with quadratic utility, $\mathbb{E}[u'(y^i(s))] = u'(\bar{y})$, so $R = 1/\beta$
- CRRA with $\gamma = 2$: $R = 0.923$, so $r = -7.7\%$ versus **4.2%** with complete markets
- A stronger desire to save need not produce more saving in equilibrium: here it produces a **lower interest rate**

The two economies side by side

	Complete markets	Bond only
Date-0 consumption c_0^i	1.00	1.00
HH1 consumption in A	1.00	1.20
HH1 consumption in B	1.00	0.80
Claims traded	∓ 0.20 per state	none
Net interest rate r	**4.2%**	**−7.7%**

- Same endowments, same preferences: the market structure alone changes consumption and the interest rate
- Welfare: the **certainty equivalent** c_{CE} is the sure consumption that gives the same utility as the lottery

$$u(c_{\mathrm{CE}}) = \tfrac{1}{2}u(1.2) + \tfrac{1}{2}u(0.8) \quad \text{with } u(c) = -\tfrac{1}{c}: \quad -\frac{1}{c_{\mathrm{CE}}} = -\frac{25}{24} \implies c_{\mathrm{CE}} = 0.96$$

- Insurance pays 1 tomorrow minus a fee f; the largest fee the household accepts solves $1 - f = 0.96$
  - at $f = 0.04$ the household is indifferent; below it, it strictly prefers insurance; above it, the lottery
- The household would surrender up to **4%** of its guaranteed insured consumption to eliminate the risk

What the small economy cannot show

- With symmetric households the bond is never used: nobody self-insures because nobody has a counterparty
- **Self-insurance** needs time: save after good shocks, draw down after bad ones
  - one date of risk is not enough to see it
- Next: one consumer, infinite horizon, prices taken as given (partial equilibrium)
  - the same two market structures: Arrow securities (Lucas and Stokey 1983) and a riskless bond (Hall 1978)
  - the same comparison: consumption and asset paths for one income history

Lucas and Stokey 1983; Hall 1978

From two dates to repeated income risk

When income risk continues indefinitely, how do consumption and assets evolve with complete markets versus a bond alone?

Two-household example	Infinite-horizon example
Two dates	Income arrives every period, indefinitely
Two households, market clearing	One household facing **externally given prices**
One future income lottery	Income follows a **Markov chain**
CRRA utility, with prudence	Quadratic utility, **without** prudence

- Partial equilibrium: the household's holdings need not be zero; it trades with a market outside the model
- Markov chain: tomorrow's income probabilities depend on today's state, $P_{ij} = \Pr(s_{t+1} = j \mid s_t = i)$

$$P = \begin{bmatrix} 0.8 & 0.2 \\ 0.4 & 0.6 \end{bmatrix}, \qquad y = (2,\ 1.5), \qquad \beta = 0.96$$

- Quadratic utility $u(c) = -\frac{1}{2}(c^\star - c)^2$: risk averse, but without prudence
  - $c^\star$ is the satiation level; removing prudence isolates what the financial instruments achieve
- The household starts in state 1 with debt of 3 ($a_0 = -3$)

Complete markets: budget and smoothing

The Arrow-security budget of the two-household example, now repeated every period:

$$\underbrace{c_t}_{\text{consume today}} + \underbrace{\sum_j q(j \mid s_t)\, a_{t+1}(j)}_{\text{buy tomorrow's claims}} = \underbrace{y(s_t)}_{\text{income today}} + \underbrace{a_t}_{\text{payment from earlier claims}}$$

- New feature: a_t, the claim bought yesterday pays out today, while a new set of claims is bought for tomorrow
  - negative a_t is a payment owed today; negative purchases raise money today by promising future payments
- Prices: $q(j \mid i) = \beta P_{ij}$, an **assumption** about market prices; in the two-household example it was an outcome
- The first-order condition for the claim on state j is the one we have seen; substitute the price:

$$q(j \mid i)\, u'(c_t) = \beta P_{ij}\, u'\big(c_{t+1}(j)\big) \implies u'(c_t) = u'\big(c_{t+1}(j)\big) \implies c_{t+1}(j) = c_t \quad \text{for either } j$$

- At these prices the household can arrange **complete smoothing**; its budget determines the level it can afford
  - at other prices complete markets alone do not give constant consumption

Complete markets: borrowing with insurance

The same agreement is renewed each period.

- "Tomorrow I repay 3 if my income is high, or 2.188 if it is low"
- "Today you lend me 2.724 if my income is high, or 2.412 if it is low": each promise priced at $q(j \mid s) = \beta P_{sj}$

$$\text{loan}(s) = \beta\left[P_{s1} \cdot 3 + P_{s2} \cdot 2.188\right]: \qquad 0.96\,(0.8 \cdot 3 + 0.2 \cdot 2.188) = 2.724, \qquad 0.96\,(0.4 \cdot 3 + 0.6 \cdot 2.188) = 2.412$$

Today's cash flows	High today	Low today
Income	$+2.000$	$+1.500$
Repay yesterday's agreement	-3.000	-2.188
Sell promise to repay if high tomorrow	$+2.304$	$+1.152$
Sell promise to repay if low tomorrow	$+0.420$	$+1.260$
Available for consumption	**1.724**	**1.724**

- Consumption is 1.724 in every state; insurance takes the form of **lower repayment when income is low**
- A run of bad shocks leaves the agreement unchanged: the household refinances, no extra debt accumulates

Amounts rounded. Asset notation: $a(1) = -3$ and $a(2) \approx -2.188$ are debts owed.

More details

Incomplete markets: a riskless bond only

The consumer can only buy a claim paying one unit tomorrow in every state, at price $1/R = \beta$:

$$c_t \;+\; \beta\, a_{t+1} \;=\; y(s_t) \;+\; a_t$$

- Euler equation with $\beta R = 1$, no binding borrowing constraint, and quadratic utility $u'(c) = c^\star - c$:

$$u'(c_t) = \mathbb{E}_t\big[u'(c_{t+1})\big] \implies c^\star - c_t = c^\star - \mathbb{E}_t\, c_{t+1} \implies \mathbb{E}_t\, c_{t+1} \;=\; c_t$$

- Today's consumption is the best forecast of tomorrow's: consumption is a **martingale** (Hall 1978)
- The household uses today's information about future income to choose consumption and saving
- Only news moves consumption: $c_{t+1} = c_t + \varepsilon_{t+1}$ with $\mathbb{E}_t\, \varepsilon_{t+1} = 0$
  - ε_{t+1} is the revision to consumption when income news arrives

Hall 1978

Incomplete markets: the consumption function

Iterate the budget forward and use $\mathbb{E}_t c_{t+j} = c_t$:

$$a_t + \mathbb{E}_t \sum_{j \geq 0} \beta^j y_{t+j} \;=\; \mathbb{E}_t \sum_{j \geq 0} \beta^j c_{t+j} \;=\; \frac{c_t}{1-\beta}$$

- Define $v(s) = \mathbb{E}\big[\sum_j \beta^j y_{t+j} \mid s_t = s\big]$, the present value of income from state s; with two states $v = (I - \beta P)^{-1} y$

$$c_t \;=\; (1-\beta)\,\big[\,a_t + v(s_t)\,\big]$$

- Consume the **annuity value** of total wealth: financial wealth a_t plus human wealth $v(s_t)$
  - consumption depends on **lifetime resources**, not on today's income alone
  - more assets, less debt, or better income prospects (higher v) all raise consumption today
  - a one-off gain of 1 raises consumption by $1 - \beta = 0.04$ at once; the rest finances future consumption
- Numbers: $v = (46.1,\ 45.3)$, annuity values $(1-\beta)v = (1.84,\ 1.81)$
  - both below 2 and above 1.5: in state 1 income exceeds its annuity value, in state 2 it falls short
  - income is 2 today, but the household owes 3 and may earn less later: it consumes 1.724, not 2
- **History matters**: earlier bad shocks leave more debt, so consumption stays lower even after income recovers

Incomplete markets: asset dynamics

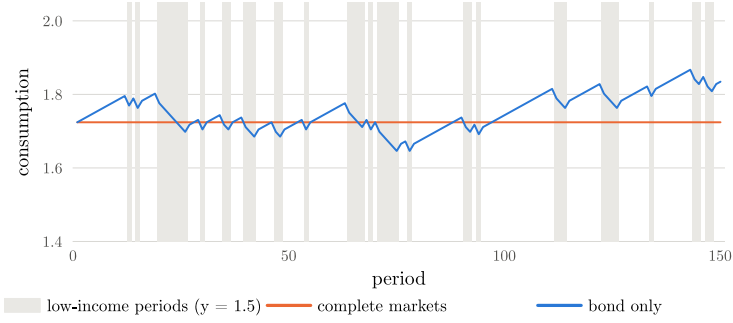
Substitute the consumption function into the budget constraint:

$$a_{t+1} - a_t \;=\; R\left[\, y(s_t) - (1-\beta)\, v(s_t) \,\right]$$

- Save when income exceeds its annuity value, dissave when it falls short
- Numbers: assets rise by **0.16** in every period spent in state 1 and fall by **0.32** in state 2
- The change depends only on today's state, not on a_t: assets have a **unit root**
  - a long run of state 2 drains assets, and consumption falls with them

Consumption paths for one income history

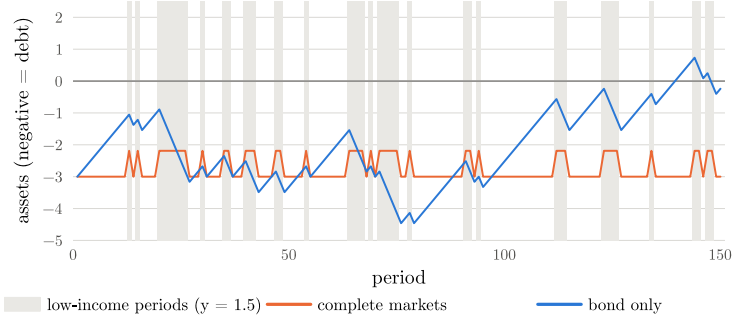
Same income draws in both economies, 150 periods: $\beta = 0.96$, $y = (2,\ 1.5)$, $P_{11} = 0.8$, $P_{22} = 0.6$, $a_0 = -3$, $s_0 = 1$. Shaded: periods with $y = 1.5$.



- Complete markets: consumption is flat at $\bar{c}$ through every income swing
- Bond only: a martingale, smoother than income, falling in shaded periods and rising outside them

Asset paths for the same income history

Complete markets: $a(1) = -3$, $a(2) = -2.19$. Bond only: $+0.16$ per unshaded period, -0.32 per shaded period.



- Complete markets: assets take two values, one per state, and never drift
- Bond only: a unit-root process; a run of low income leaves a permanent mark
- Ex-ante identical consumers with different histories end up with different wealth: **ex-post heterogeneity**

Where we stand so far?

In the infinite horizon setting, consumers lack:

- Prudence (utility is quadratic), a borrowing limit, and $\beta R < 1$
- As a result assets wander without bound: there is no **stationary wealth distribution** to study

Part 2 changes three things.

1. Restore CRRA utility and the **prudence** channel already seen in the two-household economy
2. A borrowing limit ϕ that can bind: a second precautionary motive, even without prudence
3. R set by market clearing, with $\beta R < 1$: the $2 \times 2 \times 2$ economy showed why
   - together these keep assets bounded and deliver a non-degenerate stationary distribution

Road map: three building blocks of the Bewley model.

1. The household problem, giving the **asset supply** function
   - solve the household problem, find the stationary distribution, aggregate decisions over it
2. The **asset demand** function: from an aggregate production function (firms) or the government
3. The **equilibrium** in the asset market

Model

Individual's problem

- Discrete time, infinite horizon; future utility discounted by $\beta \in (0, 1)$
- A continuum of individuals with unit mass
- Earnings are $w_t\, y_t$: market wage w_t times an idiosyncratic **labour endowment** y_t, a Markov chain with transition probabilities

$$P_{jk} \;=\; \Pr\left(y_{t+1} = y_k \,\middle|\, y_t = y_j\right)$$

- Labour is supplied inelastically; per-period utility $u(c_t)$ with $u' > 0$, $u'' < 0$, $c_t \geq 0$

Individual's problem

- Agents only have access to a **riskless bond** paying interest r; write $R = 1 + r$
  - no state-contingent Arrow securities: this is the **incomplete market**
- Saving and borrowing allowed, subject to a **borrowing constraint** ϕ (so $\underline{a} = -\phi$)
  - we assume $\beta R < 1$ and verify later that this is the only equilibrium
- Full individual problem:

$$\max_{\{c_t,\, a_{t+1}\}} \mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t u(c_t) \quad \text{s.t.} \quad c_t + a_{t+1} = w y_t + R a_t, \qquad \underbrace{a_{t+1} \geq -\phi}_{\text{borrowing constraint}} \quad , \quad c_t \geq 0,$$

  for $t = 0, 1, \ldots,$ with a_0 given

- We look for a **stationary** equilibrium, so time subscripts on prices (w, r) are dropped for now (more on that later)

Why does the borrowing constraint matter?

Two households, two periods, the same lifetime income: zero initial assets, $r = 0$, no discounting, concave u.

Income	Today	Tomorrow	Lifetime
Household A	20	100	120
Household B	60	60	120

- **Unrestricted borrowing**: both consume 60 in each period; A borrows 40 today and repays it tomorrow
- **No borrowing** ($\phi = 0$): B still consumes 60 today, but A can consume only 20
  - A would like to consume more today and less tomorrow, but cannot pledge tomorrow's income
- The same lifetime resources need not imply the same consumption today
- Future income cannot finance today's spending when borrowing is restricted

Next. How much borrowing is feasible, when the limit binds, and what binding does to optimality

Why save or borrow? A recap

1. **Intertemporal substitution**: how attractive is postponing consumption?
   - Patience β and the return R determine the reward for waiting
2. **Consumption smoothing**: how should consumption be spread across dates?
   - Save for predictable income declines; borrow against predictable income increases
3. **Precautionary saving**: how much extra should be saved because income is uncertain?
   - Compare with certain future income at the same mean
   - We have seen the prudence channel; a future borrowing constraint can add another

- The first two motives exist without income risk; the third is induced by uncertainty
- Now: when does a borrowing constraint bind, and how does that possibility affect saving?

How much debt can the household sustain?

How negative can a_t be while the household stays solvent under the **worst** income sequence?

- Iterate the budget constraint forward with $a_{t+T}/R^T \to 0$: the **intertemporal budget constraint**

$$a_t = \underbrace{\frac{1}{R}\sum_{j=0}^{\infty}\frac{c_{t+j}}{R^j}}_{\text{discounted lifetime consumption}} - \underbrace{\frac{1}{R}\sum_{j=0}^{\infty}\frac{w\,y_{t+j}}{R^j}}_{\text{discounted lifetime income}}$$

- Consumption cannot be negative, so $a_t \geq -\frac{1}{R}\sum_j w y_{t+j}/R^j$: debt cannot exceed discounted lifetime income
- A debt affordable along a favourable income sequence may be unaffordable along a bad one
- The **natural debt limit** requires solvency even under the worst sequence, $y_{t+j} = y_{\min}$ forever
  - the sum is then a geometric series, convergent since $R = 1 + r > 1$

$$a_t \geq -\frac{w\,y_{\min}}{R}\underbrace{\left(1 + \frac{1}{R} + \frac{1}{R^2} + \cdots\right)}_{=1/(1-1/R)} = -\frac{w\,y_{\min}}{R-1} = -\frac{w\,y_{\min}}{r}$$

- Debt can never exceed the discounted value of minimum income received forever

Will the household borrow up to the limit?

At the natural limit, a bad draw leaves nothing to consume.

- Enter the period with $a_t = -wy_{\min}/r$, receive $y_{\min}$, and split the budget into interest and new borrowing:

$$c_t \;=\; wy_{\min} + Ra_t - a_{t+1} \;=\; \underbrace{wy_{\min} + ra_t}_{=\,0 \text{ at the limit}} + \underbrace{(a_t - a_{t+1})}_{\leq\, 0:\; a_{t+1} \geq a_t} \;\Longrightarrow\; c_t = 0, \quad a_{t+1} = a_t$$

- Income exactly covers the interest; raising more debt is impossible because a_t is already at the limit

With $u'(c) \to \infty$ as $c \to 0$, the household stays away from it.

- When the low-income state can occur next period, the optimal choice stays strictly above the natural limit

A tighter imposed limit $\phi < wy_{\min}/r$ can still bind.

- Example: $wy_{\min} = 15$, $r = 10\%$. Natural limit $-15/0.10 = -150$: interest of 15 exhausts income, $c_t = 0$
- A lender imposes $a_{t+1} \geq -100$ instead. A household that already owes 100 has $c_t + a_{t+1} = 15 + 1.1(-100) = -95$
  - keep the debt at 100: consume 5; cut it to 95: consume 0; raise it to 105: consume 10, **not allowed**
- At the imposed limit, keeping debt unchanged allows **positive consumption**
- The household may want to consume more by borrowing further, but the constraint prevents it
  - it can therefore choose to **sit exactly at this limit**: the binding case we study next

What must an optimal choice look like?

Given assets and income today, should the household consume more now, save more, or borrow as much as permitted?

- Before computing the choice, we establish what it must satisfy. Lagrangian with multipliers λ_t and μ_t:

$$\mathcal{L} \;=\; \mathbb{E}_0 \sum_{t=0}^{\infty} \Big\{ \beta^t u(c_t) \;+\; \lambda_t \big(w y_t + R a_t - c_t - a_{t+1} \big) \;+\; \mu_t \big(a_{t+1} + \phi \big) \Big\}$$

- $\mu_t \geq 0$ is the multiplier on the **borrowing constraint**: the value of being allowed to borrow a little more at t
- Complementary slackness, $\mu_t(a_{t+1} + \phi) = 0$: extra borrowing permission is valuable only once all of it is used
- The first-order conditions combine into the **Euler equation**:

$$u'(c_t) \;=\; \beta R \, \mathbb{E}_t \big[u'(c_{t+1}) \big] \;+\; \mu_t$$

- If the constraint does not bind, $a_{t+1} > -\phi \Rightarrow \mu_t = 0$: the standard Euler equation
  - a unit saved costs utility today and yields an expected discounted benefit tomorrow
  - saving adjusts until the two balance: no marginal gain from shifting consumption across dates

When the balance cannot be struck

- If the borrowing constraint **binds**: $a_{t+1} = -\phi$ and $\mu_t > 0$, so

$$u'(c_t) \;>\; \beta R\, \mathbb{E}_t\big[u'(c_{t+1})\big]$$

- It would gain from consuming more today and repaying tomorrow, but has used all permitted borrowing
- Consumption is then fixed by the budget: all of income, assets and permitted borrowing

$$c_t \;=\; w y_t + R a_t + \phi$$

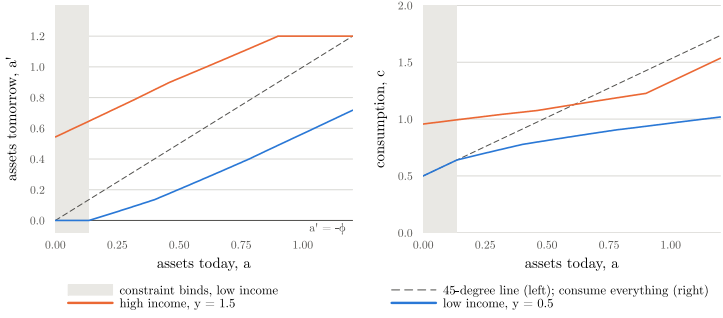
- While the household stays at the limit, an extra pound of income raises consumption by one pound: **MPC** $= 1$
  - household A: income today rises from 20 to 21, consumption rises from 20 to 21
  - the same holds if current and future income both rise, as long as the constraint still binds
  - it already wants more consumption today, so it spends the extra pound rather than saving it
- This happens when the household is poor (low wealth or income) and/or the constraint is tight: next figure

Aiyagari 1994

Where the constraint binds

- What remains: for every pair (a_t, y_t), find a_{t+1}, c_t and whether the constraint binds: the **policy function**
  - the conditions above characterise it but do not deliver it; below it is computed by value function iteration

No borrowing ($\phi = 0$), $\beta = 0.93$, $R = 1.03$, CRRA $\gamma = 2$, iid income $y \in \{0.5, 1.5\}$.



- Low income and low wealth: the household consumes everything, $a' = 0$ and $c = y + Ra$ (shaded band)
- Outside the band the policy rises with a; with high income the household saves at every wealth level
- The band widens with a tighter limit, lower βR or a larger gap between the income states

After Aiyagari 1994, Figure Ia

Saving against future borrowing constraints

We have seen a household constrained today. An unconstrained one may save more: it could be constrained tomorrow.

Suppose the household is unconstrained today: $\mu_t = 0$.

$$u'(c_t) \;=\; \beta R\, \mathbb{E}_t\big[u'(c_{t+1})\big]$$

- A bad income draw tomorrow may leave it wanting to borrow more than permitted
- In that event consumption is low: extra assets carried into the period are valuable
- Saving today builds a buffer against that possibility, even though today's constraint is slack

A second channel for precautionary saving.

- A constraint that may bind further in the future can also affect saving today
- This channel can operate even with quadratic utility ($u''' = 0$)
- Its strength depends on the borrowing limit and the size and persistence of income shocks
- In Aiyagari, prudent preferences and future borrowing restrictions both shape the policy function

Consumption-savings

- The full consumption-savings problem is solved with standard **dynamic programming** techniques
- The Bellman equation:

$$V(a, y_j) \;=\; \max_{a' \geq -\phi} \left\{ u\left(Ra + wy_j - a'\right) \;+\; \beta \sum_k P_{jk}\, V(a', y_k) \right\}$$

- Associated policy function $a' = a'(a, y)$; consumption $c = c(a, y)$ is recovered from the budget constraint
- Like Aiyagari: if y is iid, a **cash-on-hand** formulation is also available

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From partial to general equilibrium

- So far, w and r were **taken as given**: we solved the **partial equilibrium** problem of the consumer
- Next: **general equilibrium**: find the r such that the asset market clears
- Focus on the **stationary equilibrium**: aggregates (such as total assets) and prices constant through time

Sources

- Part 1: QuantEcon, *Consumption and Tax Smoothing with Complete and Incomplete Markets* (Julia lectures)
  - julia.quantecon.org/dynamic_programming/smoothing.html
  - Lucas and Stokey (1983), *Journal of Monetary Economics*; Hall (1978), *Journal of Political Economy*
  - the $2 \times 2 \times 2$ economy and the figures are course calculations in asset units, $a = -b$
- Part 2: Tomas R. Martinez (Insper), *Bewley–Huggett–Aiyagari–Imrohoroglu Model*, slides 3–20
  - Aiyagari (1994), *Quarterly Journal of Economics* 109(3), 659–684

Backup: finding the affordable consumption level

With $a_{t+1}(j) = a(j)$ the budget in each state reads $\bar{c} + \beta \sum_j P_{ij}\, a(j) = y(i) + a(i)$:

$$\bar{c} + 0.96\,[0.8\,a(1) + 0.2\,a(2)] = 2 + a(1)$$
$$\bar{c} + 0.96\,[0.4\,a(1) + 0.6\,a(2)] = 1.5 + a(2)$$

- Two equations, three unknowns $\bar{c}$, $a(1)$, $a(2)$
- Initial condition: the date-0 budget is the state-1 equation with a_0 in place of $a(1)$, so $a(1) = a_0 = -3$
- The state-2 equation then gives $a(2) \approx -2.188$, and the state-1 equation gives $\bar{c} \approx 1.724$
- Check: both budgets hold with these values, so the constant-claims plan is feasible

◂ Back

Backup: precautionary saving in two periods

Use the earlier bond notation: household i chooses the date-1 payoff a^i at price $1/R$.

- Hold $\beta R = 1$ fixed and assume the borrowing constraint is slack
- Certain date-1 income $\bar{y}$: the Euler equation gives

$$u'\left(\underbrace{\bar{y} - a^i/R}_{c_0^i}\right) = u'\left(\underbrace{\bar{y} + a^i}_{c_1^i}\right) \quad \Longrightarrow \quad a^i = 0, \qquad c_0^i = c_1^i = \bar{y}$$

- Replace certain income by the original lottery, $\mathbb{E}[y^i(s)] = \bar{y}$
- At the new optimum, with prudence and nondegenerate income risk:

$$u'(c_0^i) = \mathbb{E}[u'(c_1^i(s))] \; > \; u'(\mathbb{E}[c_1^i(s)]) \quad \Longrightarrow \quad c_0^i < \mathbb{E}[c_1^i(s)]$$

- Since $c_0^i = \bar{y} - a^i/R$ and $\mathbb{E}[c_1^i(s)] = \bar{y} + a^i$, this requires $a^i > 0$
- Risk raises desired saving at the fixed benchmark return; market clearing must then change R

Back

Backup: Jensen's inequality

- A function $f(x)$ is **convex** if, for $\alpha \in [0,1]$ and any two points x_1, x_2:

$$f\left(\alpha x_1 + (1-\alpha)x_2\right) \;\leq\; \alpha f(x_1) + (1-\alpha) f(x_2)$$

- Take a random variable x equal to x_1 with probability p and x_2 with probability $1-p$:
  - expected value: $\mathbb{E}[x] = px_1 + (1-p)x_2$
  - by convexity of f: $f(\mathbb{E}[x]) \leq \mathbb{E}[f(x)]$
- The general case of this result is **Jensen's inequality**
- In our problem: whether $\mathbb{E}[u'(c_1)] \gtrless u'(\mathbb{E}[c_1])$ depends on the convexity of $u'(\cdot)$

◂ Back